

BOOKLET I

SECTION A: STRUCTURED QUESTIONS

QUESTION 1

The cell wall of the prokaryote *Escherichia coli* is made of peptidoglycan that consists of polysaccharide chains cross-linked by short chains of amino acids. **Fig. 1.1** shows that the polysaccharide chain is made up of two different sugars N-acetylmuramate (NAM) and N-acetylglucosamine (NAG), which alternate along the chain.

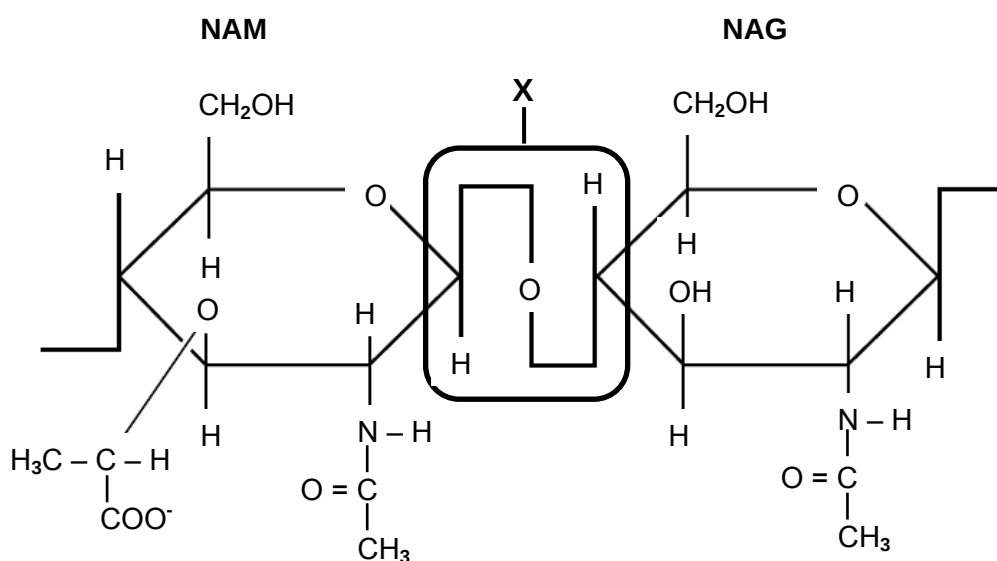


Fig. 1.1

- (a) State the name of the bond represented by X. [1]

- (b) Distinguish between bacterial peptidoglycan and cellulose found in the plant cell wall. [2]

- (c) Suggest why peptidoglycan is cross-linked with amino acids. [1]

Bacteria **A** and **B** have different outer layers, as shown in **Fig. 1.2**. Bacterium **A** is susceptible to the antibiotic penicillin while bacterium **B** is not.

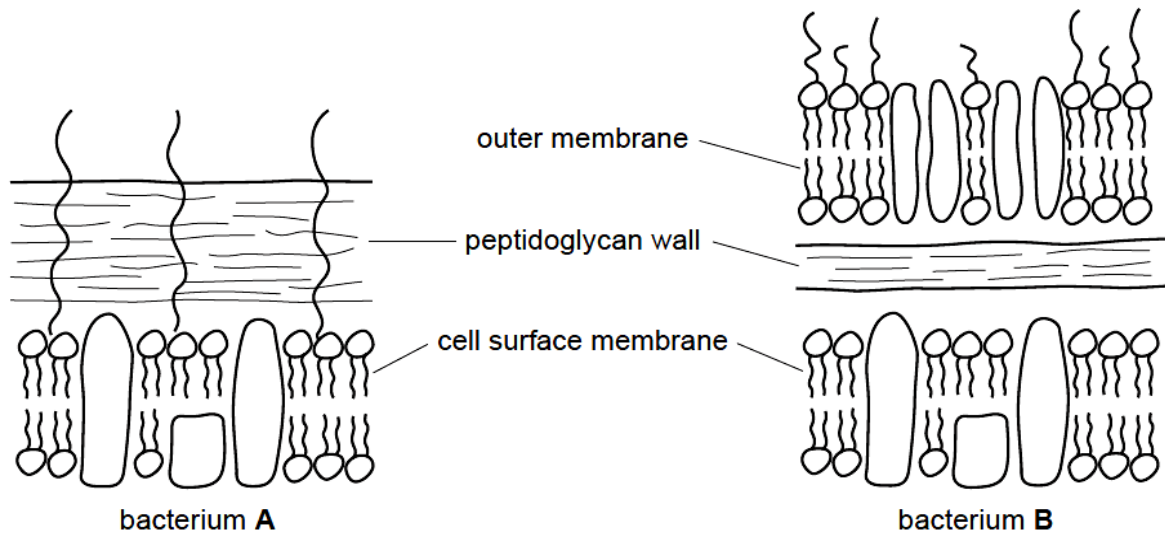


Fig. 1.2

(d) With reference to **Fig. 1.2**,

- (i) describe how the cell surface membrane in bacterium **B** is structurally different from the peptidoglycan wall. [2]

- (ii) suggest why bacterium **A** is susceptible to penicillin while bacterium **B** is not. [1]

Metabolic pathways can be controlled by end-product inhibition of enzyme-catalysed reactions. KAS III is the initial enzyme of fatty acid production in bacteria. The substrate for this reaction is malonyl-ACP.

Three different mutant strains of bacteria were generated, each with a different mutated *KAS III* gene: **M1**, **M2** and **M3**. The enzyme activity of the wild-type and the three mutant strains was tested with and without the addition of a KAS III inhibitor, dodecanoyl-ACP. Dodecanoyl-ACP is structurally similar to malonyl-ACP. **Fig. 1.3** shows the mean activity of KAS III in the wild-type, **M1**, **M2** and **M3** strains.

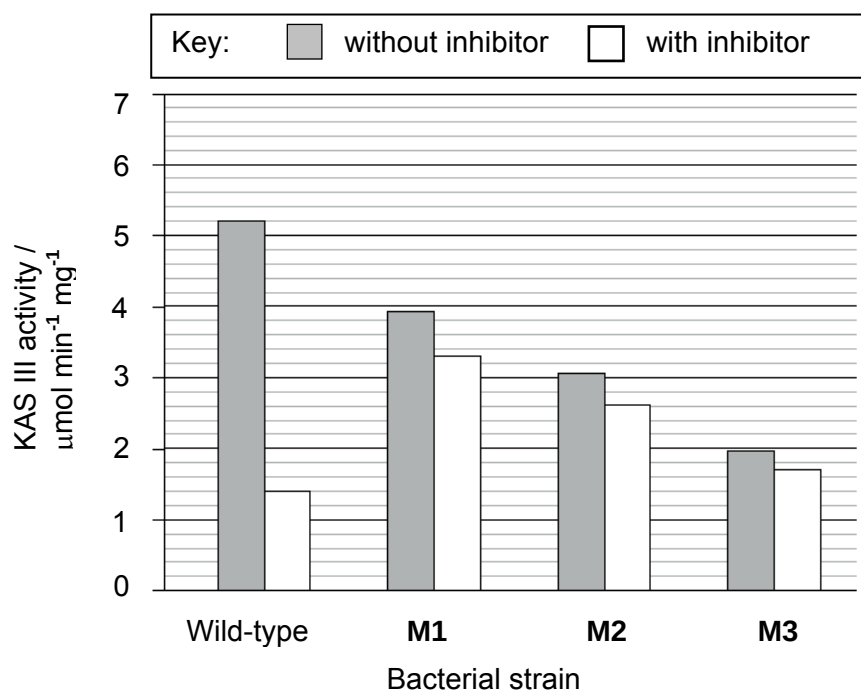


Fig. 1.3

- (e) With reference to **Fig. 1.3**, explain why the activity of KAS III in the wild-type strain changes when dodecanoyl-ACP is added. [2]

- (f) Suggest an explanation for the effect of mutation on KAS III activity. [2]

Fig. 1.4 shows the effect of increasing malonyl-ACP concentration on the rate of reaction catalysed by KAS III in the wild-type strain.

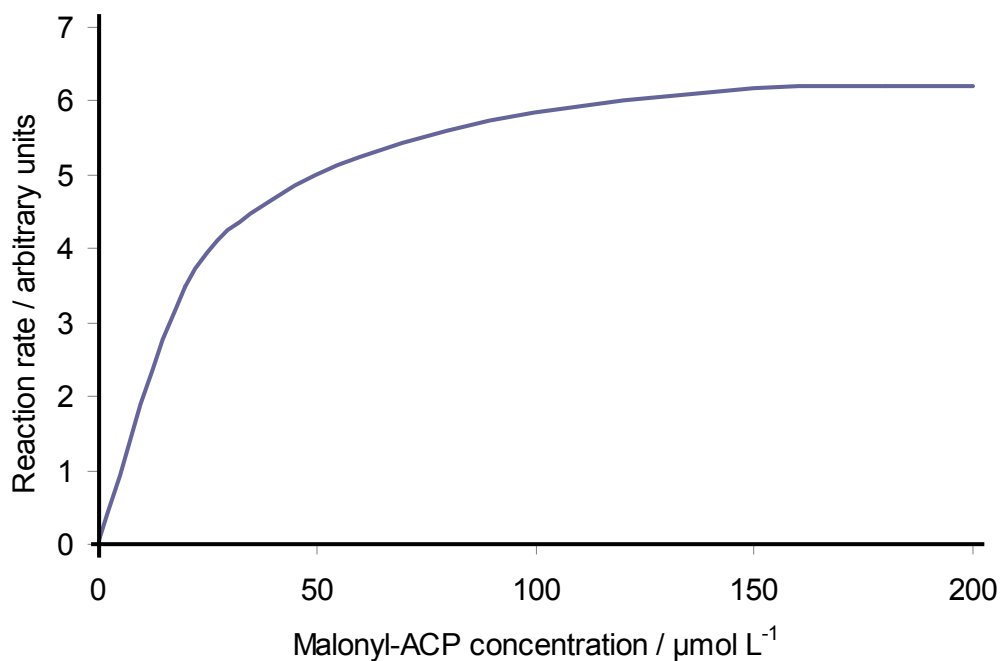


Fig. 1.4

- (g) Using information from **Fig. 1.3**, sketch a curve in **Fig. 1.4** to represent the effect of the addition of dodecanoyl-ACP on the rate of reaction with increasing malonyl-ACP concentration. [1]

[Total: 12 marks]

QUESTION 2

(a) Draw a fully labelled diagram of a 3-bp long DNA molecule, using the space provided below.

[3]

In the fruit fly *Drosophila melanogaster*, changes to its DNA occur during DNA replication, resulting in increased expression of affected genes. **Fig. 2.1** shows one way by which this effect arises for chorion genes involved in the development of egg cells.

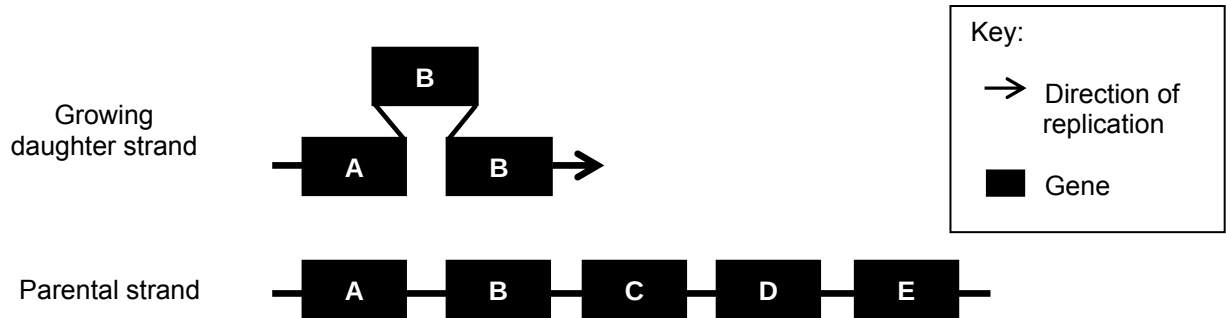


Fig. 2.1

(b) With reference to **Fig. 2.1**,

(i) explain how increased gene expression is achieved in *Drosophila melanogaster*. [2]

(ii) suggest how the mechanism shown can result in gene deletion in the newly synthesised daughter strand. [1]

Transcription of the chorion gene sequences produces RNA transcripts that must be successfully processed before they can be exported out of the nucleus into the cytoplasm for translation, as shown in **Fig. 2.2**.

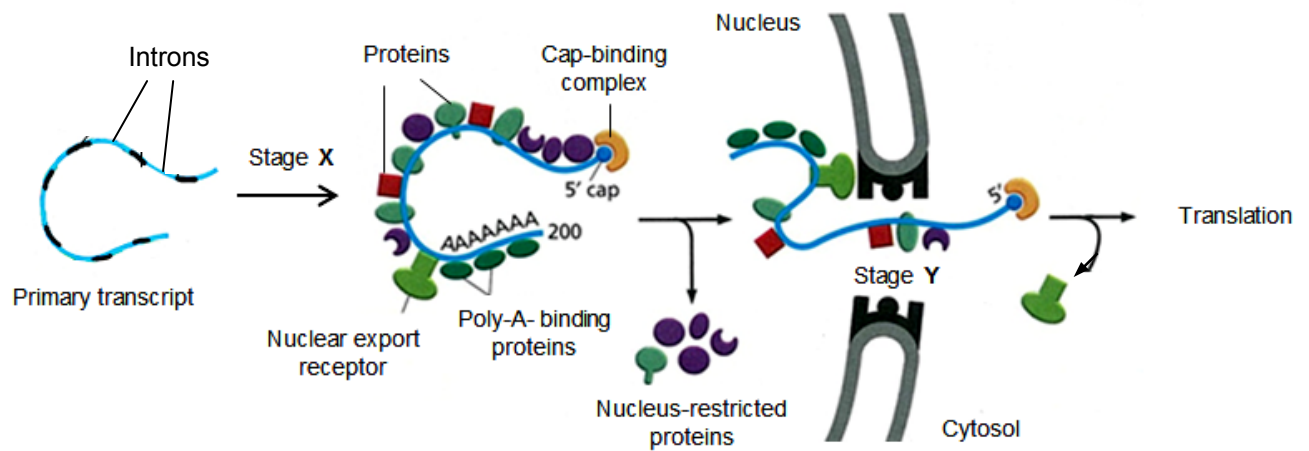


Fig. 2.2

(c) With reference to **Fig. 2.2**,

- (i) describe one process occurring at stage X that results in the formation of a continuous mRNA coding sequence. [2]

- (ii) suggest why the cap-binding complex and poly-A-binding proteins are essential for stage Y. [1]

There are distinct differences between eukaryotic and prokaryotic translation.

(d) Outline two such differences. [2]

[Total: 11 marks]

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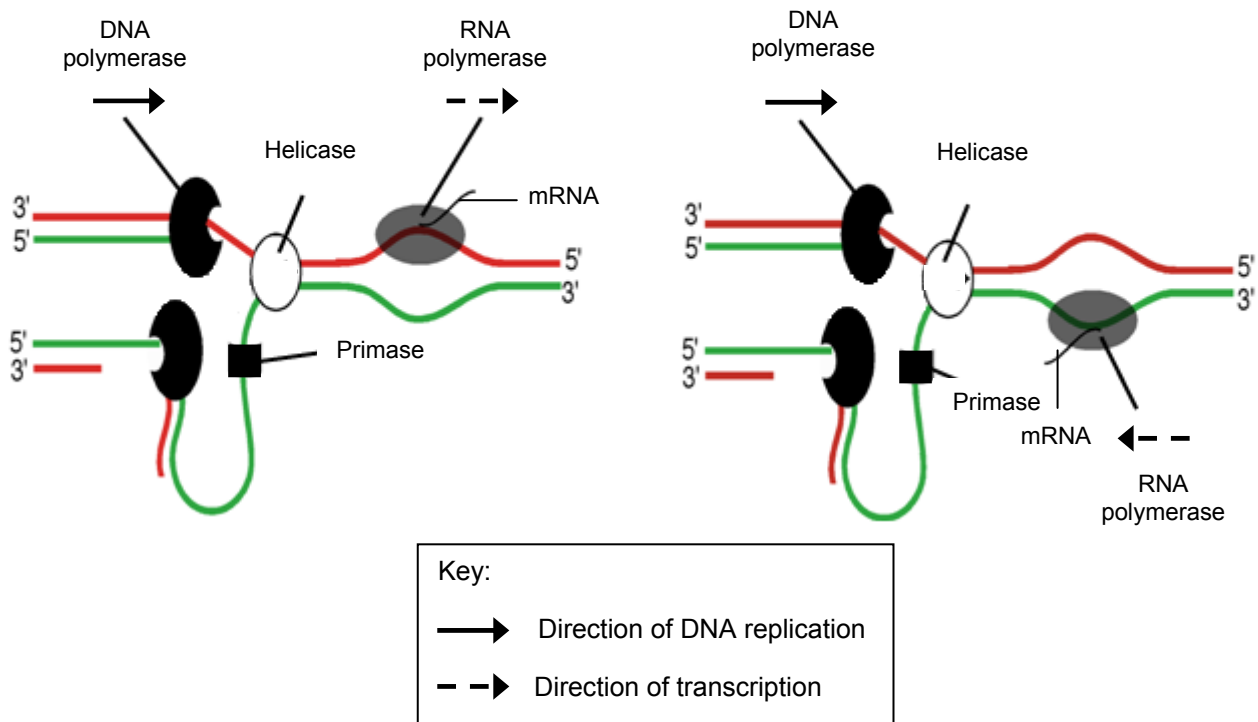
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BOOKLET II

SECTION A: STRUCTURED QUESTIONS (continued)

QUESTION 3

Prokaryotic chromosomal replication often takes place simultaneously with transcription. Collisions between DNA polymerases (estimated replication rate of 500 bp / sec) and RNA polymerases (estimated transcription rate of 50 bp / sec) are thus inevitable. Co-oriented and head-on collisions may occur as shown in **Fig. 3.1**, leading to decreased rates of, and even aborted, replication and transcription. As a result, highly expressed genes tend to be organised near the origin of replication.



Co-oriented collision resulting in decreased rates of replication and transcription

Head-on collision resulting in aborted replication and transcription

Fig. 3.1

- (a) Explain why prokaryotic chromosomal replication often takes place simultaneously with transcription. [2]

- (b) Explain how these collisions can occur between DNA and RNA polymerases. [3]

Bacterial genes are organised into operons, such as the *trp* operon. The effect of the presence of tryptophan on the *trp* repressor is shown in Fig. 3.2.

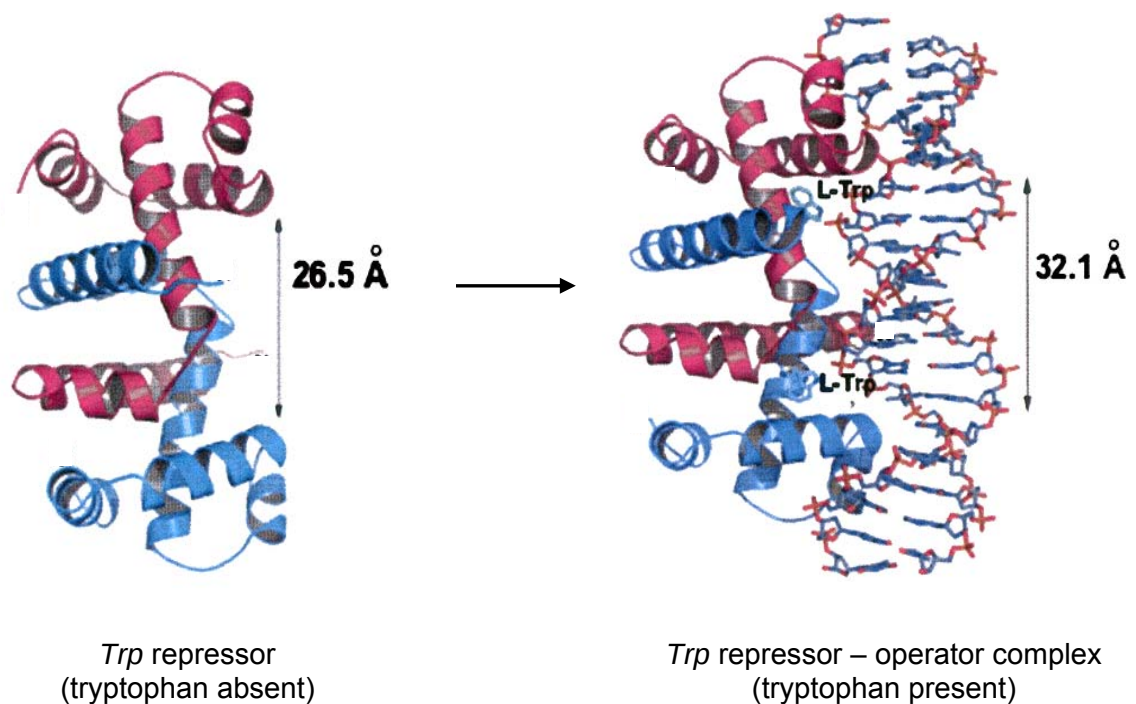


Fig. 3.2

- (c) Describe what happens to the *trp* repressor when tryptophan is present. [3]

- (d) Describe the significance of transcriptional regulation through repression of the operon. [2]

Studies have shown that the *trp* operon contains three operators.

(e) Suggest the purpose of having more than one operator. [1]

[Total: 11 marks]

QUESTION 4

Human papillomavirus (HPV) infection and cigarette smoking could lead to head and neck cancer. This cancer develops in the surface epithelial cells lining the oral and nasal cavities as well as the throat. **Fig. 4.1** shows how HPV infection and cigarette smoking triggers the transformation of normal tissue into a malignant tumour as the cancer progresses. **E6** and **E7** are viral proteins and Rb is the gene product of a tumour suppressor gene.

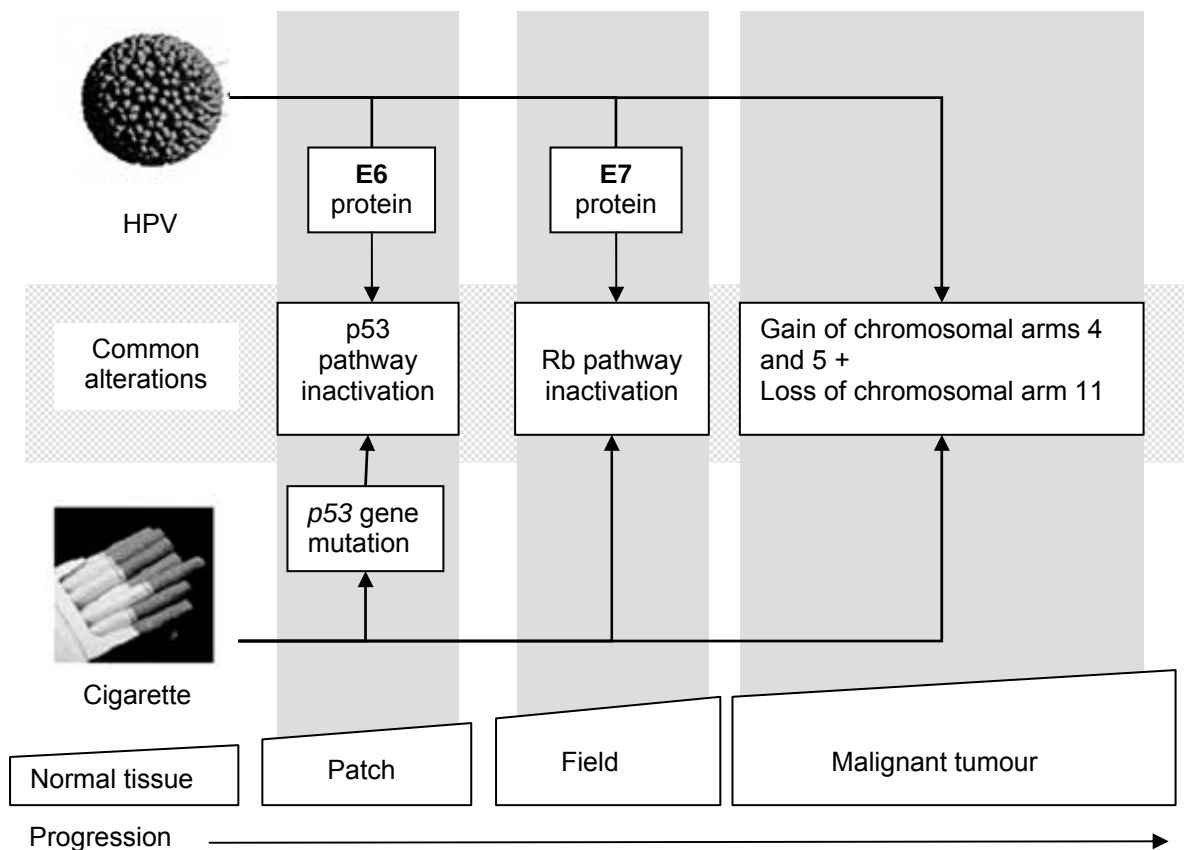


Fig. 4.1

(a) Describe how cigarette smoking could lead to *p53* gene mutation.

[3]

- (b) Explain how **Fig. 4.1** provides evidence for the development of cancer as a multi-step process. [2]

Fig. 4.2 shows how the **E7** protein can interact with the Rb protein.

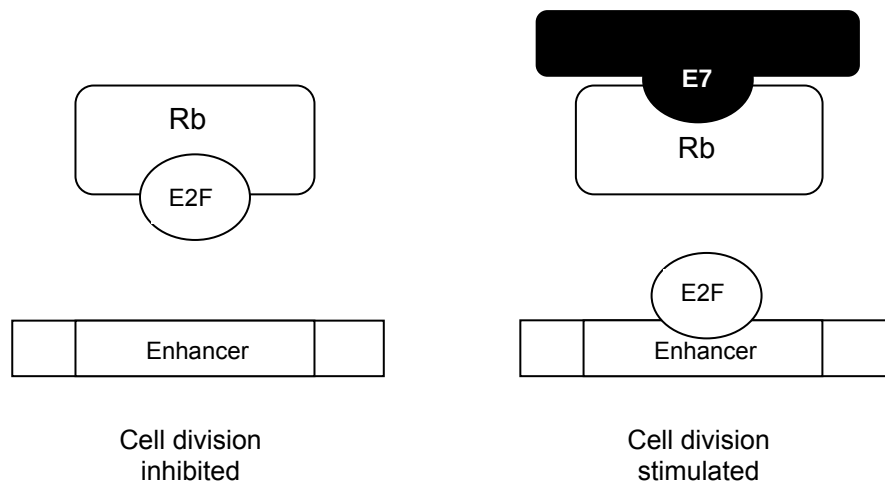


Fig. 4.2

- (c) With reference to **Fig 4.1** and **Fig. 4.2**, explain how the **E6** and **E7** proteins could lead to eventual changes in chromosomal structure in malignant tumour cells. [4]

- (d) Explain one consequence of gain-of-function mutation in both alleles of a proto-oncogene in the germline of a man. [2]

[Total: 11 marks]

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BOOKLET III

SECTION A: STRUCTURED QUESTIONS (continued)

QUESTION 5

Glucose-6-phosphate dehydrogenase (G6PD) deficiency is a sex-linked recessive condition.

- (a) Define what is meant by a *sex-linked* condition. [1]

A normal Indian man marries a normal woman, whose father has G6PD deficiency. The couple decides to have a child.

- (b) State the probability that their first boy will be affected with G6PD deficiency. Give a reason for your answer. [2]

Both the Indian man and his wife have blood group AB. However, their first boy has blood group O. Upon consultation with a geneticist, they realised that this is due to the boy lacking the H antigen. This antigen is an essential precursor of the ABO blood group antigens. The boy is homozygous recessive at the H gene locus on chromosome 19. Hence, he cannot synthesise either the A or B antigen on his red blood cells even if he may have the I^A or I^B allele at the ABO gene locus on chromosome 9.

- (c) Using suitable symbols, draw a genetic diagram in the space provided below to show the expected blood group phenotypic ratio in the offspring of the Indian couple. [5]

The geneticist carried out a research on this unique blood phenotype of the boy, known as the Bombay phenotype, and found out that it is a very rare condition present in about 0.0004% (about 4 per million) of the human population. He wanted to verify its occurrence rate and did an investigation on 200 couples where both partners have blood group AB, just like the Indian couple. The blood groups of their children are shown as follows:

95 children with blood group A

168 children with blood group AB

97 children with blood group B

The expected phenotypic ratio of a cross between a couple where both partners are homozygous dominant at the H gene locus on chromosome 19 and have the same blood group AB is 1 blood group A: 2 blood group AB: 1 blood group B.

The chi-square (χ^2) test is used to determine if the results of this investigation are in accordance with the expected results. The formula for χ^2 and the table of probabilities are given as follows:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Degrees of freedom	Probability		
	0.10	0.05	0.01
1	2.71	3.84	6.64
2	4.69	5.99	9.21
3	6.25	7.82	11.35
4	7.78	9.49	13.28

- (d) (i)** Using the information provided, calculate the χ^2 value for the observed results. Show your working clearly in the space provided below. [2]

- (ii) Deduce if the observed results follow the expected phenotypic ratio of 1 blood group A: 2 blood group AB: 1 blood group B. Explain your answer. [2]

[Total: 12 marks]

QUESTION 6

In response to the rise in blood glucose after a carbohydrate-rich meal, a series of cellular molecular events including event **X** occur, as shown in **Fig. 6.1**.

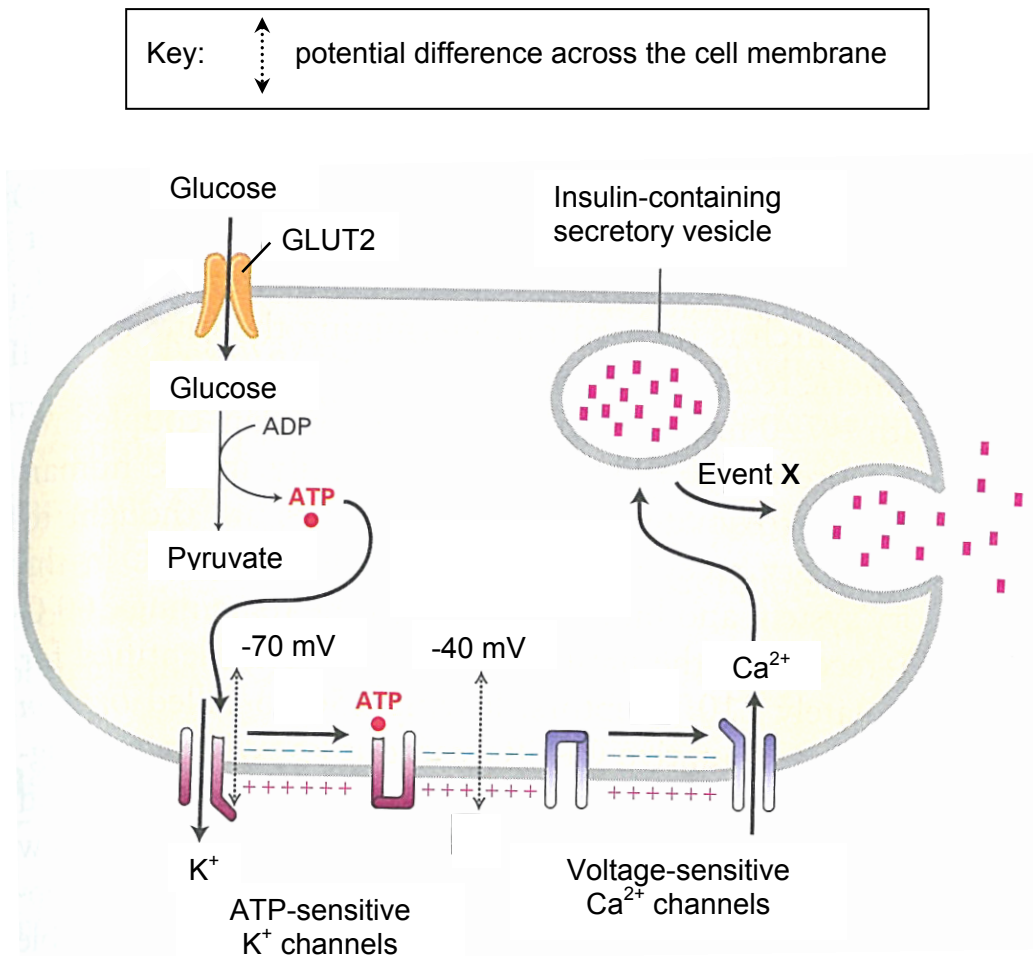


Fig. 6.1

(a) With reference to **Fig. 6.1**,

(i) identify the cell type.

[1]

(ii) state a piece of visible evidence to support your answer to (a)(i).

[1]

(iii) suggest how the binding of ATP to ATP-sensitive K^+ channels leads to an influx of Ca^{2+} ions. [3]

(b) Suggest how the influx of Ca^{2+} ions leads to event X. [2]

Fig. 6.2 shows the normal changes in the blood glucose concentration of an individual over time in hours (h) with the oral intake of small, successive doses of glucose. The concentration of blood glucose prior to the first dose was steady at $100 \text{ mg } 100 \text{ cm}^{-3}$.

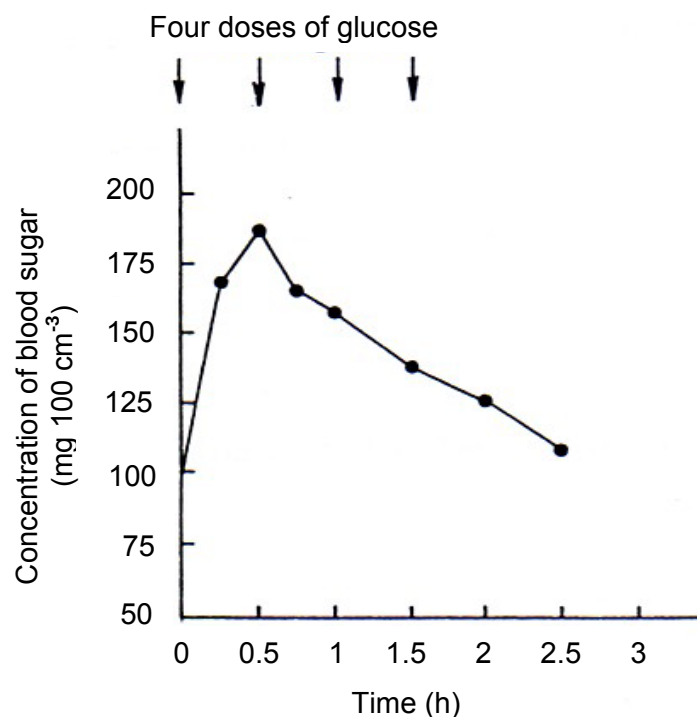


Fig. 6.2

- (c) With reference to **Fig. 6.1** and **Fig. 6.2**, explain why the second dose of glucose does not have the same effect as the first. [3]

- (d) Suggest why an injection of insulin into its target cells would not produce the same changes in the individual's blood glucose concentration as represented in **Fig. 6.2**. [1]

[Total: 11 marks]

BOOKLET IV

SECTION A: STRUCTURED QUESTIONS (continued)

QUESTION 7

Variation is important for populations of organisms to evolve.

(a) Explain why the population is the smallest unit that can evolve.

[2]

Evolution of populations may lead to speciation. Polyploidy is one way by which speciation can occur. Observed in both plants and animals, polyploidy is well represented in rainbow trout and modern bread wheat.

Triploid rainbow trout *Oncorhynchus mykiss* may be produced by fertilising eggs in which the meiotic cycle is inhibited by heat-shock. A diploid egg nucleus then combines with a haploid sperm nucleus to produce a triploid zygote. **Fig. 7.1** shows the growth of diploid female rainbow trout and triploid female rainbow trout in a reservoir over a four-year period.

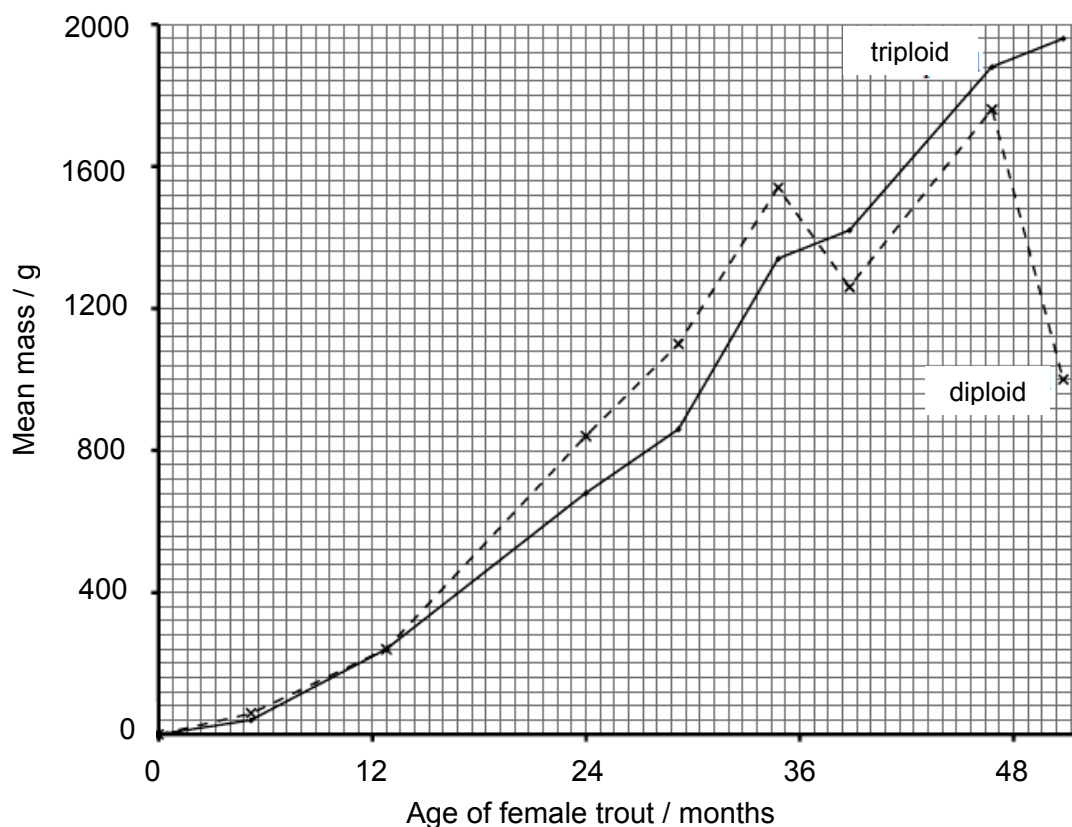


Fig. 7.1

(b) (i) Describe the trends shown in **Fig. 7.1**.

[2]

All triploid fish are sterile and female triploids exhibit restricted ovary development so that eggs are not produced. Triploids are also generally less aggressive (e.g. in their feeding) than diploids. Diploid trout in this reservoir become sexually mature after their third year, and female diploids produce hundreds of eggs.

(ii) Using the information provided, suggest explanations for the trends identified in **Fig. 7.1**. [2]

Modern polyploid varieties of wheat have developed from numerous hybridisation events between different species of wild grass. **Fig. 7.2** shows some of the possible steps that are believed to have been involved in the development of bread wheat *Triticum aestivum*. The letters **A**, **B** and **C** represent three different sets of seven chromosomes.

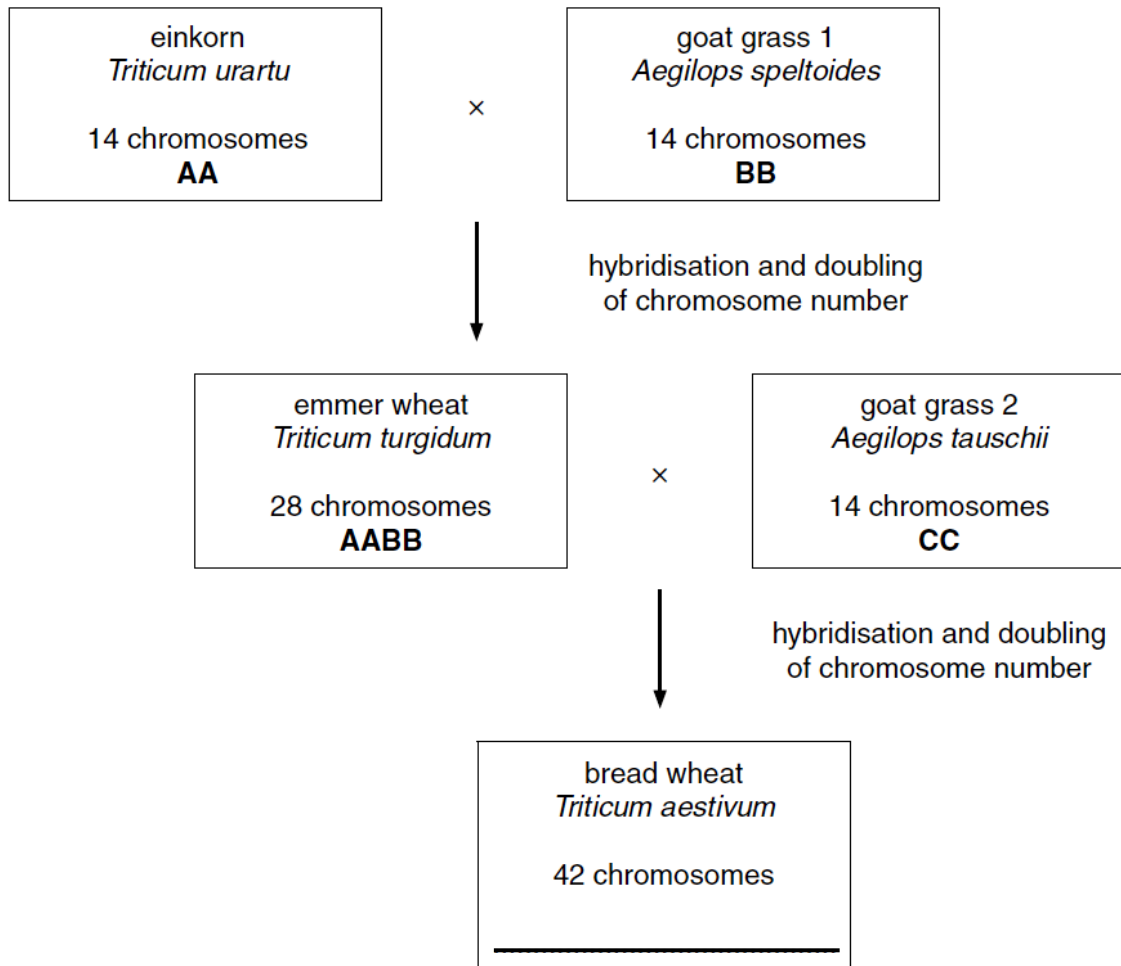


Fig. 7.2

(c) (i) Complete **Fig. 7.2** by writing letters on the line provided to represent the sets of chromosomes in bread wheat. [1]

(ii) Explain why hybridisation between emmer wheat and goat grass 2 would have produced a sterile hybrid, if doubling of chromosome number had not occurred. [2]

- (iii) With reference to **Fig. 7.2**, suggest why *Triticum urartu* and *Triticum turgidum* are classified as different species. [2]

- (iv) *Triticum turgidum* emerged as a new species without being geographically isolated from *Triticum urartu*. Suggest, with a reason, if this form of speciation can lead to adaptive radiation. [1]

[Total: 12 marks]

--- END OF SECTION A ---

SECTION B: FREE RESPONSE QUESTIONS

Answer **ONE** of the following two free response questions.

Your answers must be in continuous prose, where appropriate.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

BEGIN EACH PART ON A FRESH SHEET OF WRITING PAPER.

A **NIL RETURN** is required.

QUESTION 8

- (a) Describe the role of substrate-level phosphorylation in the production of ATP. [6]
- (b) With reference to photosynthesis and cellular respiration, explain the importance of compartmentalisation within a plant cell. [10]
- (c) Suggest the effects of a season of drought on the process of Calvin cycle in a plant. [4]

QUESTION 9

- (a) Describe the changes involving homologous chromosomes during meiosis, and explain how these changes ensure genetic variation and stability. [10]
- (b) Explain how the structure of components of the Human Immunodeficiency Virus (HIV) is related to their functions in the viral reproductive cycle. [6]
- (c) Suggest how viruses can cause diseases other than cancer. [4]

[Total: 20 marks]

--- END OF SECTION B ---

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